

It has become increasingly important for a successful academic community to consist of various cultural, racial, and ethnic groups, because *diversity* lends us a global lens to look into different backgrounds and mindsets and helps create a more inclusive workplace. While the United States is a country rich in multicultural communities, its universities are not immune to gender, racial, ethnic and other biases currently impacting academia.

When an on-campus-USPS employee despised me for my accent and refused to mail my documents, when a local shuttle service publicly discriminates international students in my university, and when I heard people claiming that female colleagues received good job offers because of their gender rather than their excellence, I realized how biases and prejudice are negatively influencing the way we behave towards certain people. I have also realized the urgency of building a more diversified academic community. Next, I will talk about the challenges and my tentative plans.

Challenges: There are many challenges in our way to achieve diversity in academia, particularly in engineering. Although women represent 47% of the workforce, only 12% of engineers are female.¹ African-American and Latino workers represent 29% of the workforce, but only 12% of the engineering workforce, according to the 2015 U.S. News/Raytheon STEM Index.² In my opinion, the general challenges are people's inadequate awareness of diversity issues, and lack of motivation to implement policies that promote diversity. As an AI and machine learning (ML) researcher myself, I think it is also necessary to point out that the wide use of AI & ML, which has created the so-called "filter bubbles," is also posing to us new challenges for promoting diversity.

1. **Awareness of unconscious biases.** Admittedly, significant improvements have been made in identifying and addressing conscious bias with laws and policies to prevent explicit prejudices. Unconscious bias, on the other hand, refers to unintentional but deeply ingrained stereotypes about certain groups of people. Unconscious biases are usually far more prevalent than people think. For example, I notice that there are still baby songs singing "father works late at night and mother cooks and does the dishes." People with unconscious biases tend to promote those who are like themselves, which is a big barrier to diversity.
2. **From awareness to policies.** Diversity is easier said than done. With continuous effort, more and more people have begun to think about and discuss this problem, but we need to go beyond that, to implement effective policies to address it. My wife, as a breastfeeding mom, once attended a diversity-promoting workshop and ended up without a nursing room available, even though she had requested it multiple times. Diversity problems were being discussed in the workshop, but we need more comprehensive policies to show that diversity is not just a slogan, but a commitment that we should make.
3. **New challenges in the AI & ML era.** AI & ML have been the driving force of many business models, where algorithms are used to selectively show information that a user presumably likes to see. The users may get significantly less contact with different viewpoints and become intellectually isolated. Examples include Google's personalized search engine and Facebook's personalized news-stream feed. It is very common to see that female users see online commercials for makeup and household cleaning products, while male users see beers and burgers. Besides causing other obvious social problems

¹"Solving the Equation: The Variables for Women's Success in Engineering and Computing," American Association of University Women (AAUW), <https://www.aauw.org/research/solving-the-equation/>

²"To attract more blacks and Hispanics to STEM, universities must address racial issues on campus," The Hechinger Report, <https://hechingerreport.org/attract-blacks-hispanics-stem-universities-must-address-racial-issues-campus/>

that need to be addressed, these so-called “filter bubbles” have also posed new challenges to diversity because they perpetuate stereotypes and create separation among different groups of people.

Plan: My experience has made me realize the importance of advocating diversity in academia, especially in engineering. Based on my understanding of the challenges, I propose the following plan to promote diversity.

1. **Make AI & ML a solution to, rather than a source of, diversity problems.** According to an article on *The Guardian*, “more than 60% of Facebook users are entirely unaware of any curation on Facebook at all, believing instead that every single story from their friends and followed pages appeared in their news feed.”³ First, it is important to raise public awareness of the new type of diversity problems that AI & ML are causing. Second, as educators, we need to teach the next generation of students these potential problems in the filter bubble era. Third, while social media seek to please users and make a profit, we researchers need to understand the biases in these techniques and find ways to correct them. I am very happy to learn that many AI researchers have noticed this problem.⁴ Finally, AI & ML can also be a solution to many diversity problems, e.g., translating American Sign Language into text and detecting hate speeches online.
2. **Build students’ interest and confidence in engineering.** The unconscious biases that certain groups are not good at engineering have been strengthened by the lack of presence of these groups in a long time. To stop these biases, it is important to “counter-stereotype,” i.e., to promote underrepresented groups in engineering. The promotion is difficult because of the inequality in accessing to educational resources and the resulting lack of confidence in choosing a career in engineering for these groups. Schools in developed areas understand the importance of early exposure to engineering. However, lots of other schools may not have access to the same resource and students from those areas tend to believe that engineering is intimidating and that studying engineering is not worthwhile.

I think that it is critical to help young students clearly understand what engineering is and be aware of the potential career paths ahead of them. I plan to work with the Education Department in my university to build programs to integrate more engineering into the K-12 curricula and to train professional teachers who can help inspire the next generation to be competent and confident in engineering.

3. **Organize more career-building programs for underrepresented groups.** Diversity in education stretches further than just the diversity of students. Having diverse faculty can also help break stereotypes. I could see how much my wife, who was a Ph.D. student herself, had benefited from many career-building programs for female Ph.D. students in her career, such as the Rising Star in the Fields of EECS program and the CRA Women Grad Cohort. My wife was able to build connections, get job opportunities, and be motivated by senior professors by their own stories of success. I think these career-building programs can also extend beyond gender to other underrepresented groups. I would be more than happy to make contributions to these programs.

Promoting diversity in academia and in engineering still has a long way to go. I feel obliged to first stop unconscious biases that I may have, and then help build a more diverse and inclusive community. I hope that, with our continuous effort, people are judged by their contributions rather than their gender, race, ethnicity and many other stereotypes.

³Hern (2017-05-22). “How social media filter bubbles and algorithms influence the election”. *The Guardian*.

⁴E.g., see Sun et al., “Mitigating Gender Bias in Natural Language Processing: Literature Review,” in *ACL’19*.